PROJECT 67 / DATABASE SYSTEMS

DBMS REVIEW – 2

Gym Membership and Fitness Session Management System

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System scope, then → now

Review 2 turns the conceptual model into an enforceable, testable database.

MEMBERSHIP
Members · Plans · Subscriptions

FITNESS
Classes · Schedules · Bookings

COACHING
Trainers · Assessments · Programmes

OPERATIONS
Equipment · Maintenance · Payments

REVIEW 1

Problem · scope · users · requirements
ER model · initial relational schema

REVIEW 2

3NF · dependencies · DDL · rules
queries · views · indexing · transactions · tests

Normalization is a sequence of decisions

Each stage removes a specific source of redundancy and anomaly.

UNF

Repeating groups and mixed facts

1NF

Atomic values one fact per field

2NF

No partial dependencies

3NF

No transitive dependencies

`Example refinement`

Member details + membership plans + subscriptions

Separate entities keep identity, pricing and lifecycle facts independent.

Functional dependencies reveal the design

The determinant on the left identifies the facts on the right.

```
Member_ID → Name, Email, Phone, Date_of_Birth
```

```
Plan_ID → Plan_Name, Duration, Price
```

```
Trainer_ID → Name, Specialization, Phone
```

```
Subscription_ID → Member_ID, Plan_ID, Dates, Status
```

ANOMALIES REDUCED

UPDATE

INSERTION

DELETION

Evaluate candidate keys—not only declared primary keys.

Compact data dictionary

Representative attributes; exact types may vary by DBMS.

TABLE	ATTRIBUTE	TYPE	CONSTRAINT	DESCRIPTION
Members	Member_ID	INT	PRIMARY KEY	Unique member identifier
Members	Email	VARCHAR(100)	UNIQUE, NOT NULL	Member email address
Membership_Plans	Price	DECIMAL(10,2)	CHECK (Price >= 0)	Membership plan price
Subscriptions	Start_Date	DATE	NOT NULL	Subscription start date
Subscriptions	End_Date	DATE	NOT NULL	Subscription end date
Fitness_Classes	Capacity	INT	CHECK (Capacity > 0)	Maximum class capacity
Assessments	Weight	DECIMAL(5,2)	CHECK (Weight >= 0)	Recorded member weight
Payments	Amount	DECIMAL(10,2)	CHECK (Amount >= 0)	Payment amount

DDL makes integrity explicit

Constraints move essential rules closer to the data.

```
CREATE TABLE Membership_Plans (
  Plan_ID INT PRIMARY KEY,
  Plan_Name VARCHAR(50) NOT NULL,
  Duration_Days INT NOT NULL
    CHECK (Duration_Days > 0),
  Price DECIMAL(10,2) NOT NULL
    CHECK (Price >= 0)
);
```

PRIMARY KEY
unique identity

FOREIGN KEY
valid relationships

NOT NULL / UNIQUE
required + distinct

CHECK / DEFAULT
valid range + fallback

Joins turn relationships into reports

Member → subscription → plan is the core reporting path.




```
SELECT m.Name, p.Plan_Name, s.Status
FROM Members m
JOIN Subscriptions s
  ON m.Member_ID = s.Member_ID
JOIN Membership_Plans p
  ON s.Plan_ID = p.Plan_ID;
```

INNER JOIN
member + plan + status

LEFT JOIN
members without bookings

MULTI-TABLE
registrations + attendance

Aggregate, compare, investigate

GROUP BY creates management-level summaries; HAVING filters the summaries.

COUNT()
registrations

SUM()
successful payments

AVG()
weight · body fat

```
SELECT Member_ID, SUM(Amount) AS Total_Paid
FROM Payments
WHERE Status = 'Success'
GROUP BY Member_ID;
```

SUBQUERIES

Above-average payers

Classes with above-average registrations

PROJECT 67 · REVIEW 3

DBMS REVIEW-3

Gym Membership and Fitness Session Management System

Application Development, Testing and Final Demonstration

Student Names · Roll Numbers
Faculty Guide · Department · University



From schema to working system

Review 3 closes the loop: connected application, controlled operations and demonstrable evidence.

CENTRALIZED SCOPE



Members · subscriptions · trainers · classes · attendance

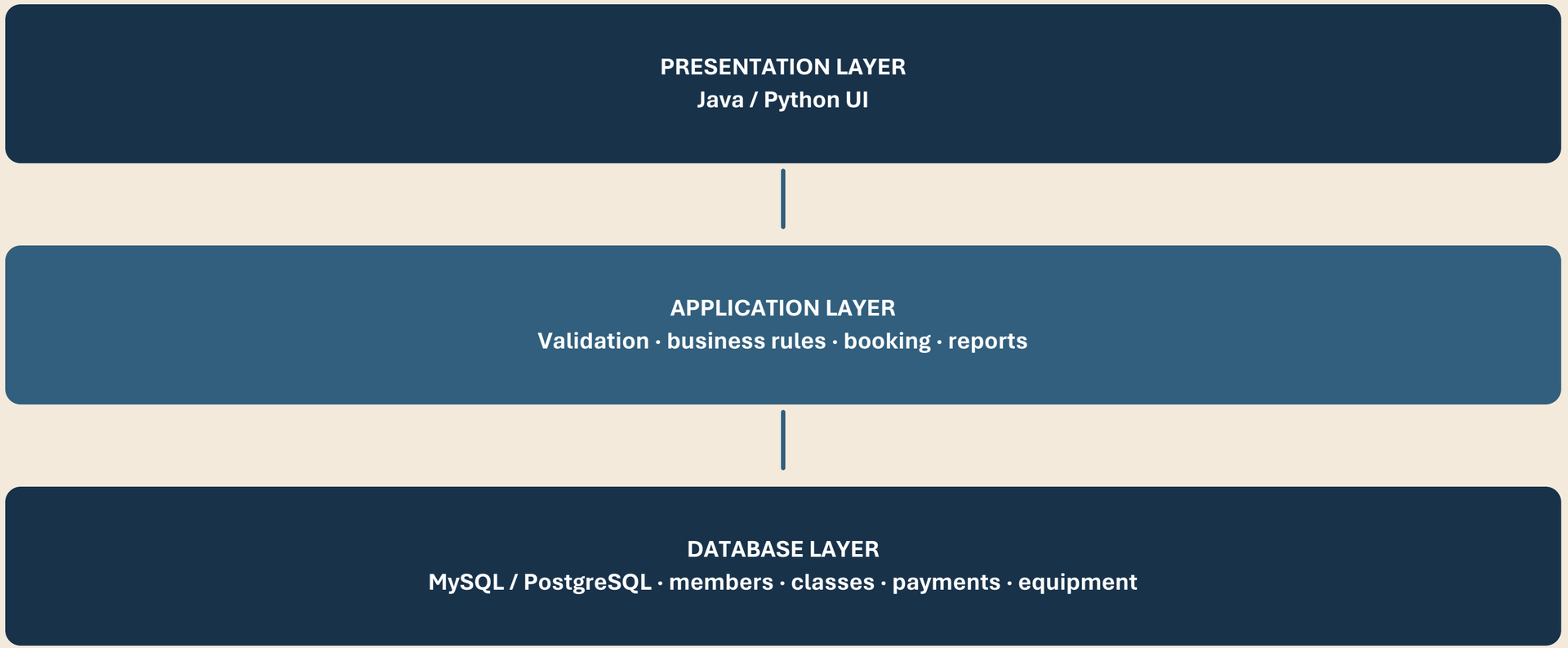
Assessments · workout plans · schedules · payments · equipment · maintenance

Review 1 — requirements, users, scope, ER model **Review 2 — 3NF, constraints, SQL** **Review 3 — application, validation, reports, testing**

Three layers. One trusted data path.

Separation of concerns keeps the interface responsive, rules consistent and records reliable.

PKs, FKs and constraints preserve integrity



USER → APPLICATION → DATABASE → QUERY RESULTS → USER INTERFACE

Technology stack & connectivity

A desktop client communicates through parameterized queries and a controlled transaction lifecycle.

FRONTEND

Java Swing / JavaFX
or Python Tkinter

BACKEND

CRUD · validation
business logic

DATABASE

MySQL / PostgreSQL

CONNECTOR

JDBC / Python DB connector



Credentials secured · graceful connection and query error handling

One application. Role-aware views.

The dashboard reveals only the actions and records each role is authorized to access.

ADMINISTRATOR

Members · plans · trainers
Schedules · equipment · reports

MEMBER MANAGEMENT

Membership · classes
Bookings · renewal status

TRAINER PORTAL

Assigned sessions
Bookings · attendance

MANAGEMENT REPORTS

Subscriptions · attendance
Revenue · maintenance

CRUD OPERATIONS

1 → 2 → 3 → 4

CREATE

Add members, trainers and classes

READ

View and search records

UPDATE

Edit member details and renew memberships

DELETE

Remove permitted records while maintaining data integrity.

EXAMPLE SQL

```
INSERT INTO Members VALUES
(101, 'Rahul', 'rahul@example.com', '9876543210');

SELECT * FROM Members;

UPDATE Members SET Name = 'Rahul K'
WHERE Member_ID = 101;

DELETE FROM Members WHERE Member_ID = 101;
```



Benefit:

Simplifies record management and maintains data consistency.

Member & membership management

A single workspace supports registration, search, plan assignment and lifecycle visibility.

MEMBER FORM

- Member ID
- Name
- Email
- Phone
- Plan
- Start Date
- End Date
- Subscription Status

MEMBERS

ID	NAME	PLAN	END DATE	STATUS
M-104	A. Sharma	Annual	14 Jun 2026	ACTIVE
M-105	N. Rao	Monthly	22 Mar 2026	EXPIRING
M-106	J. Thomas	Quarterly	02 Feb 2026	EXPIRED

CREATE → READ → UPDATE → DELETE

Illustrative application screen — replace with genuine build screenshot

Trainer assignment & fitness records

Validated assessments and conflict-aware scheduling connect people, programmes and progress.

TRAINER ASSIGNMENT

Profiles & specializations

Contact information

Assigned classes and sessions

Schedule overlap detection

Illustrative interface placeholder

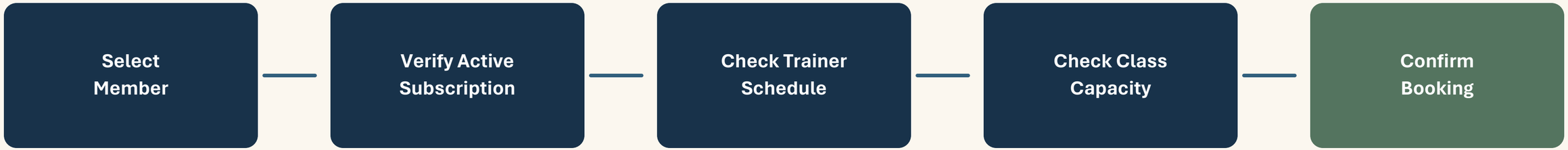
ASSESSMENT / PROGRAMME

Weight	72.5 kg
Height	178 cm
Body-fat	18.2%
Assigned trainer	[select]
Programme dates	[from — to]

VALIDATE: no negative values · valid dates · no trainer overlap

Booking is a controlled transaction

A seat is confirmed only after subscription, trainer and capacity checks all pass.



CAPACITY RULE

```
confirmed registrations < maximum capacity
```

CONCURRENCY CONTROL

transaction + database locking
prevents double booking

Any failed rule → reject safely and roll back

Attendance becomes an operational signal

Authorized trainers and administrators record status once; summaries serve multiple views.

REG ID	MEMBER	CLASS	SCHEDULE	STATUS
R-201	A. Sharma	Strength	08:00–09:00	ATTENDED
R-202	N. Rao	Yoga	09:30–10:30	ABSENT
R-203	J. Thomas	Cardio	18:00–19:00	ATTENDED
R-204	P. Sen	Strength	08:00–09:00	REGISTERED

Illustrative attendance table

SUMMARY

24
registered

18
attended

6
absent

Payments, renewals & expiry tracking

Database-led status reflects actual outcomes; renewal history preserves both old and new dates.

PAYMENT HISTORY

DATE	AMOUNT	METHOD	STATUS
04 Mar 26	2,500	Card	PAID
04 Feb 26	2,500	Transfer	FAILED
04 Jan 26	2,500	Cash	PAID

Outstanding and failed payments remain visible — no dummy success claims.

RENEW MEMBERSHIP

Member ID	M-105
Old expiry	22 Mar 2026
New expiry	22 Apr 2026
Payment status	Confirmed

Expired · expiring soon · renewed

Equipment availability, planned—not guessed

Maintenance history and next-service dates turn records into safer operating decisions.

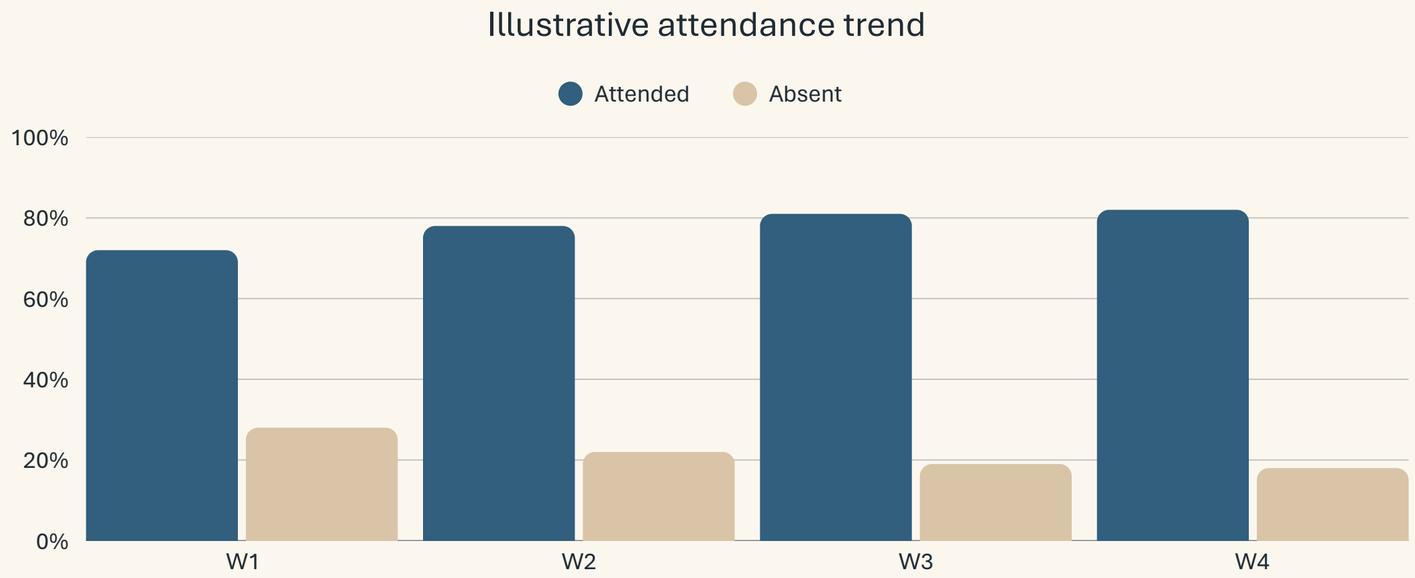
EQUIPMENT ID	NAME	STATUS	LAST MAINT.	NEXT MAINT.	COST
EQ-011	Treadmill 3	AVAILABLE	12 Jan 2026	12 Apr 2026	4,500
EQ-018	Cable Station	MAINTENANCE	28 Feb 2026	15 Mar 2026	7,800
EQ-024	Spin Bike 6	UNAVAILABLE	04 Mar 2026	10 Mar 2026	2,200

Service planning → fewer disruptions → higher equipment availability

SQL views power decision-ready reports

Illustrative dashboard values below must be replaced by actual sample-data query results.

128	14	82%	₹—
Active memberships	Upcoming renewals	Attendance rate	Revenue



MEMBERSHIP
active · expired · renewal

TRAINER
sessions · assigned classes

FINANCIAL + EQUIPMENT
revenue · maintenance cost

Implementation progress & test plan

Claim evidence only after execution; placeholders remain for screenshots and results.

- ☐ Duplicate email detection
- ☐ Foreign-key integrity
- ☐ Negative amounts / invalid dates
- ☐ Expired subscriptions / renewals
- ☐ Full-class capacity limit
- ☐ Overlapping trainer schedules

IMPLEMENTED

Tables + relationships

Constraints

Valid-key sample data

Representative queries

EVIDENCE PLACEHOLDERS

SQL execution · query results · DB tool visuals

A normalized core.
A reliable application.

3NF reduces redundancy. Constraints protect integrity. Joins, aggregates, subqueries and views turn gym operations into useful reports.

NEXT → REVIEW 3

Java or Python application for CRUD, bookings, attendance, renewals, maintenance and reporting.

THANK YOU